

Bridgewater Data Classification & AI Governance Strategy

1. Overview: Governance Challenges

Bridgewater is currently navigating a critical challenge regarding data classification within their AI agent ecosystem. This initiative is highly visible, driven by direct focus from the CEO and the Chief Compliance Officer. The primary objective is to implement a robust governance framework that ensures AI agents correctly identify, handle, and protect sensitive data types, including:

- **Personal Identifiable Information (PII):** Ensuring regulatory compliance.
- **Non-Public Information (NPI) :** Internal proprietary data protection.
- **Material Non-Public Information (MNPI):** Preventing market abuse and ensuring strict adherence to financial regulations.
- Privileged: Tag it and access
- Security classification

The core goal is to verify that AI agents are both receiving the correct information as input and generating safe, compliant responses as output.

2. Scope of the Problem

The governance framework must address data classification at three distinct stages of the AI lifecycle:

- **Input Data Classification:** Analyzing and tagging data as it enters the agent environment to determine its sensitivity levels.

- **Tool Call Governance:** Monitoring interactions between the agent and external tools, such as websearch, to ensure sensitive data is not leaked and that retrieved information is properly categorized.
- **Output Data Classification:** Scrubbing or validating agent-generated content before it is presented to a user to prevent the accidental disclosure of MNPI or PII.

3. Core Engine Objectives

The primary goal is to establish a unified framework consisting of two core functional engines:

- **Systematic Classification Engine:** Automatically categorizes and tags data at all stages, ensuring sensitivity levels are consistently identified and enforced across all workflows.
- **Verification Engine:** Acts as an independent validation layer that inspects and verifies data interactions, tool usage, and generated outputs to maintain regulatory compliance and safety standards.

4. Integration Options

Bridgewater is evaluating two primary architectural patterns for where classification logic should reside in the agent pipeline:

- **Pre-ingestion:** A static approach where data is classified and filtered before it reaches the agentic reasoning layer.
- **Agent Run-time:** A dynamic approach where classification occurs in real-time as the agent processes data and interacts with tools, allowing for more contextual governance.

5. Competitive Landscape

The following solutions and competitors are being explored by Bridgewater to address these classification needs:

Provider / Tool	Solution Focus
AWS Bedrock	Guardrails utilizing keywords and lexicons for data filtering.
Microsoft (MSFT)	Implementation of trainable classifiers for more complex data patterns.
Proofpoint	External security and data loss prevention (DLP) integration.
Affinity	Relationship and data intelligence governance.
Run Layer	Dedicated infrastructure layer for policy enforcement.

6. Internal Evaluation & Strategic Goals

The internal Google team has set the following priorities to advance this engagement:

- **Identify Support Teams:** Map out which internal Google Cloud and Security teams (e.g., Vertex AI Safety, Cloud DLP, or Mandiant) are best equipped to address Bridgewater’s specific PII and MNPI requirements.
- **Partnership Assessment:** Determine the strategic value of a formal partnership with Bridgewater to co-develop or integrate these governance capabilities into our core AI offerings.

- **Resource Allocation:** Evaluate the technical feasibility of providing a solution that outperforms the "trainable classifiers" or "lexicon-based" approaches of competitors.

with that classification embedded directly into our data foundation and enforced at every step of agent execution.

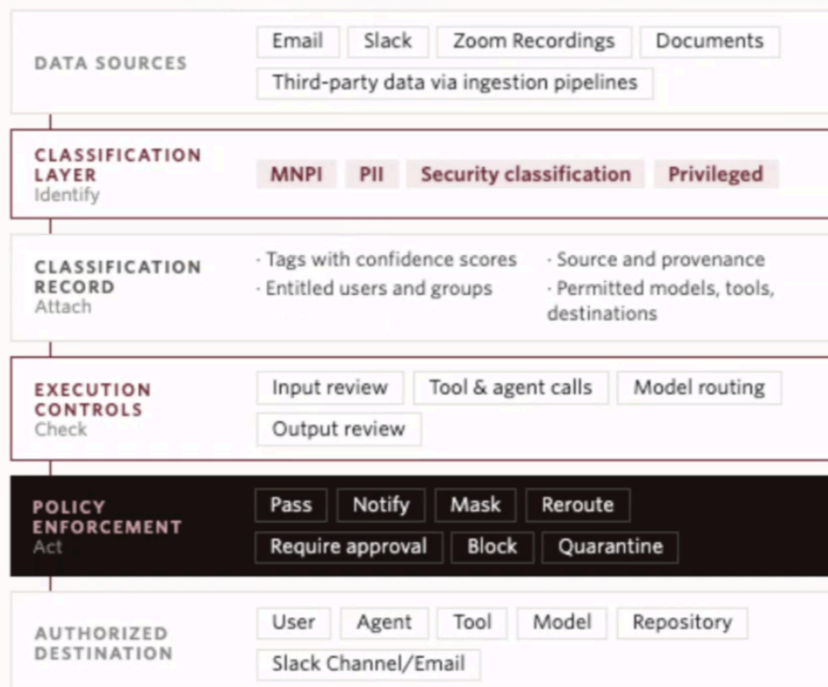
WHY THIS IS IMPORTANT

Humans — backed by strong compliance, employment, and security policy — have been the protection against misuse of sensitive data. As agents take on business workflows, that protection must be **designed into the data layer itself**. Without it, agentic use of data carries real regulatory risk; solving it later would block the tools that are central to our business strategy. Every enterprise with agentic ambitions faces this problem — we intend to help shape the solution, not wait for one.

IMMEDIATE PRIORITY & MILESTONE

- 01 Prove a **classifier we trust**: benchmark candidates against golden datasets built from historically reviewed cases.
- 02 Run it at **agent runtime** — flagging risky data to the user in the flow of work:
- 03 **Risk-reducing, not risk-eliminating** — matching the standard our controls hold today, while the full architecture is built out.

HIGH-LEVEL ARCHITECTURE — WHERE WE WANT TO HEAD



SURROUNDING LOOP

Governance & evaluation

Golden datasets

Historical cases, reviewed examples, synthetic edge cases.

Benchmark

Every classifier and candidate tool measured against the same set.

Monitor

False positives, false negatives, confidence calibration.

Update

Classifiers, policies, and ownership — with named owners and a review cadence.

WHAT WE ARE LOOKING FOR